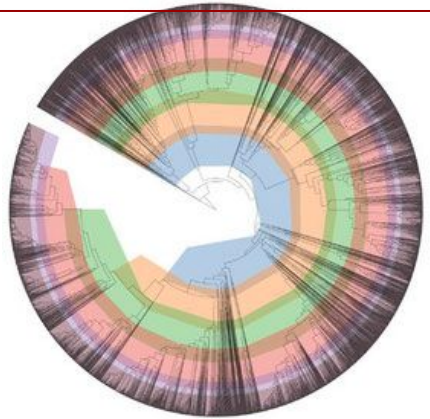
ENVIRONMENTAL METAGENOMICS

Physalia course, online, 12–16 October 2026

Gene catalog approach

Nikolay Oskolkov, Lund University, NBIS SciLifeLab
Luis Pedro Coelho, Queensland University of Technology

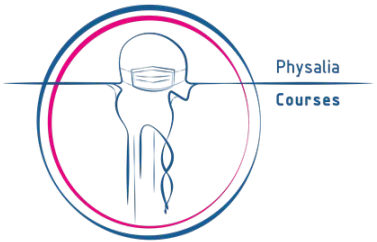


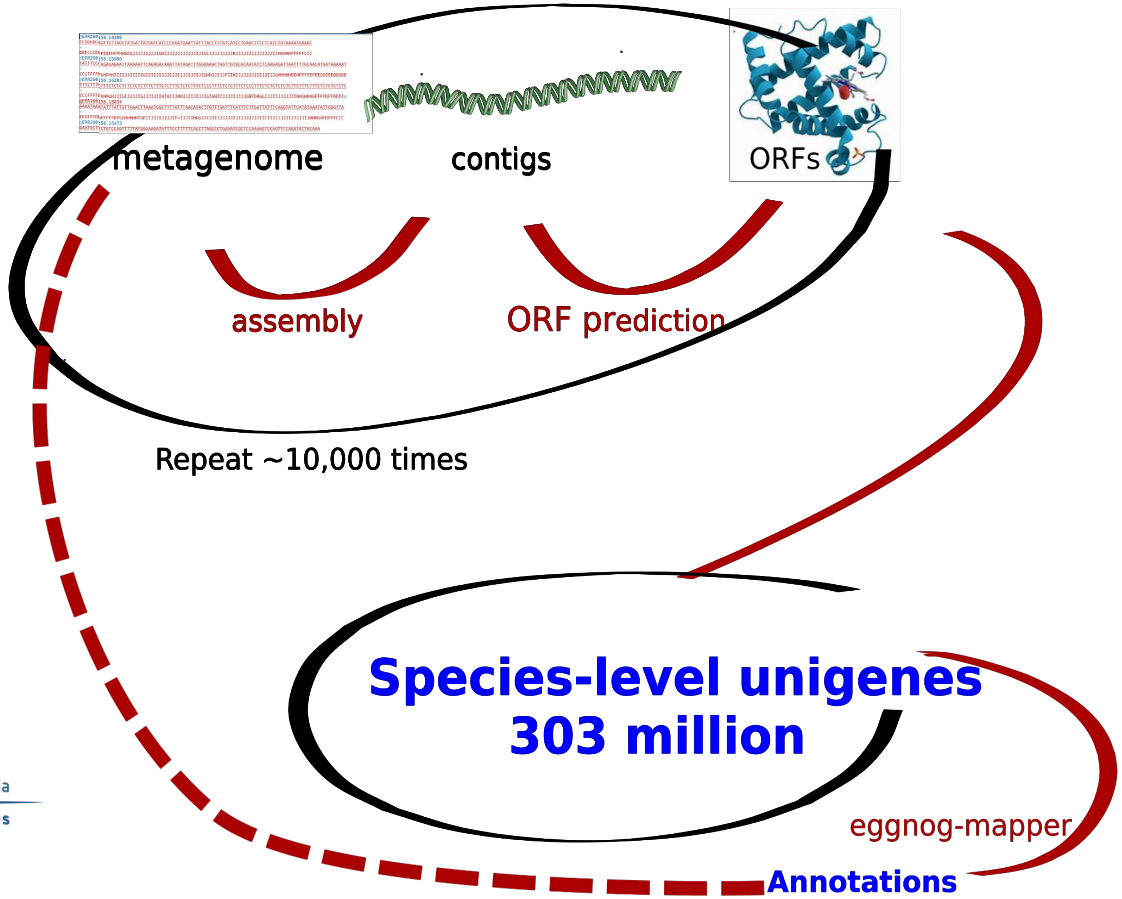
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NB: original course material courtesy:
Dr. Antti Karkman, University of Helsinki
Dr. Igor Pessi, Finnish Environment Institute (SYKE)

Gene catalogue approach

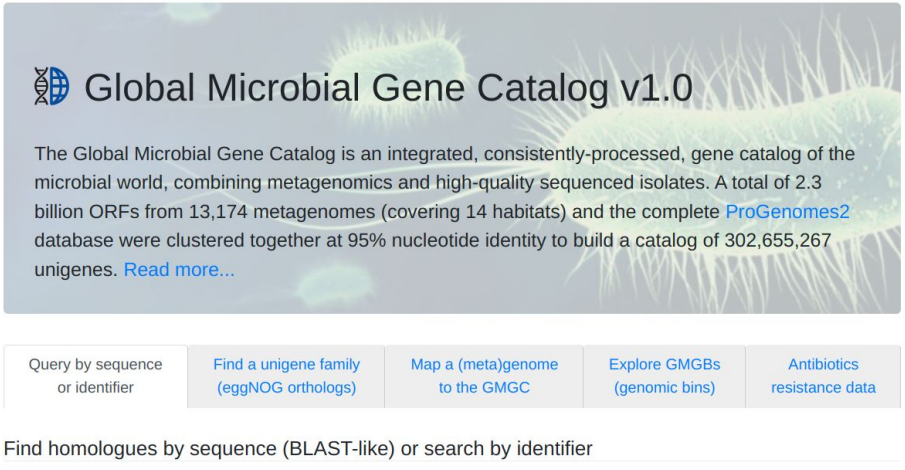
1. Take *contigs* (binned or unbinned)
2. Predict genes (ORFs)
3. Cluster all the genes together
    a. 95% nucleotide identity
4. This is your **gene catalogue**
5. Annotate it with eggnog-mapper/RGI/…
6. Align reads to the catalog

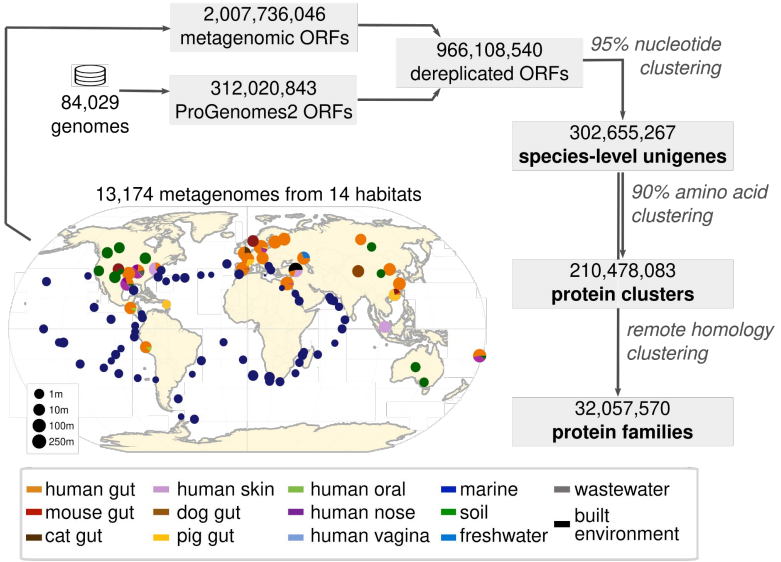




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Existing gene catalogues





From (Coelho et al., Nature, 2022)

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